

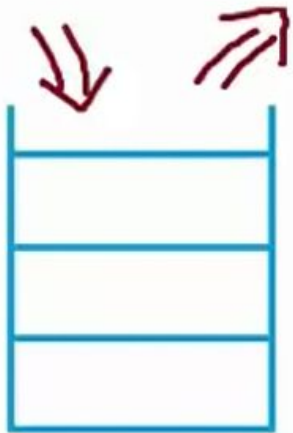
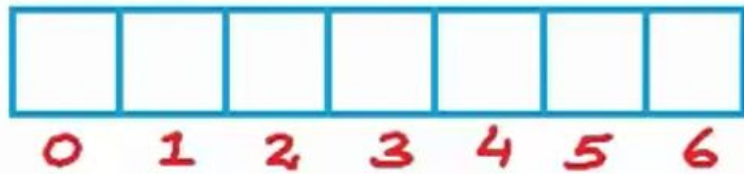
Tree



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Linear data structures

Array



Stack



Linked List



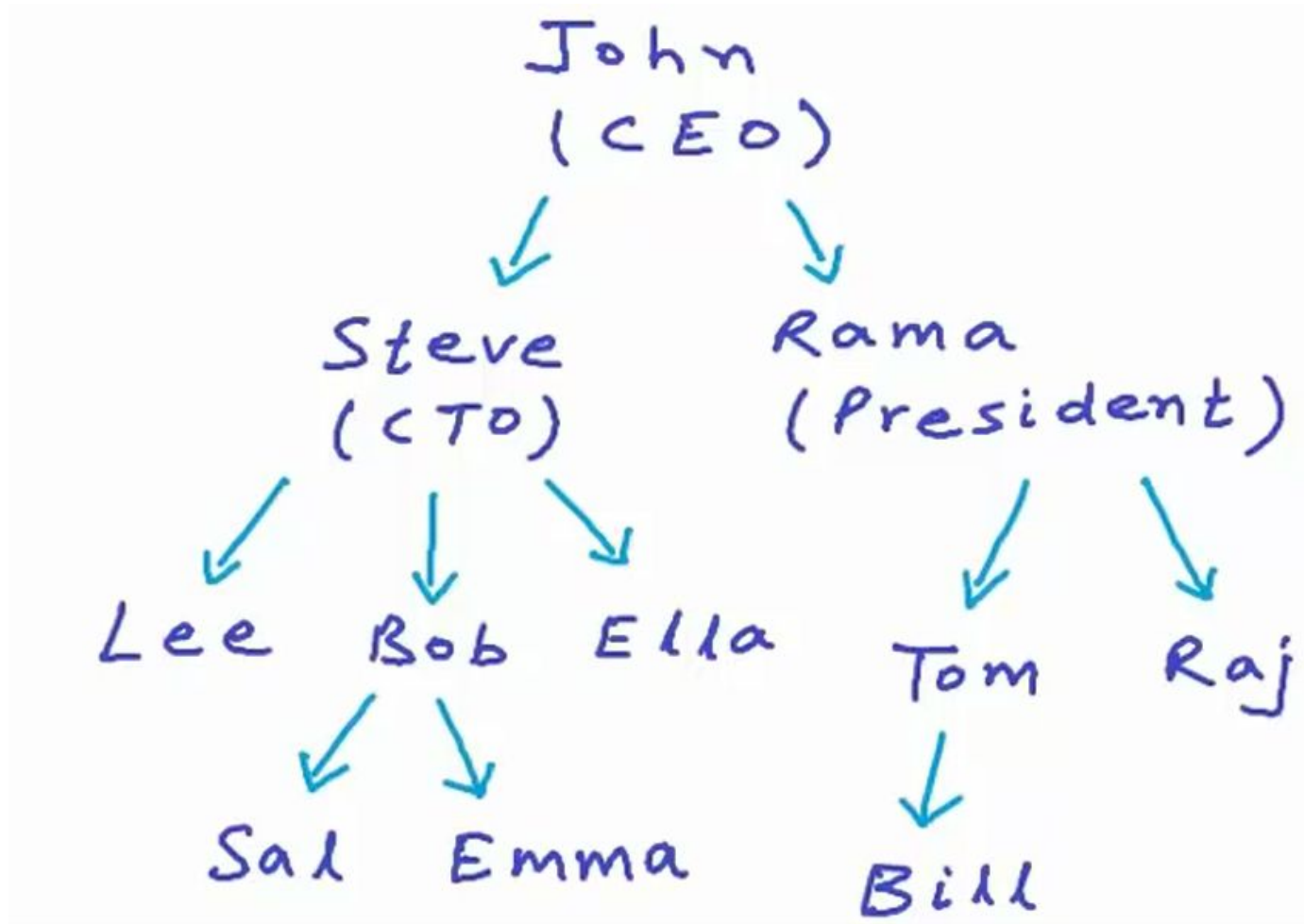
Queue

How should i decide which data structure to use?

- ❖ What needs to be stored?
- ❖ Cost of operations
- ❖ Memory usage
- ❖ Ease of implementation

What do we use a tree for?

- ❖ Trees are used for representing hierarchical data.



Tree Data Structure

1. What is a tree structure?

Ans: A tree structure is a data structure that consists of nodes connected by edges, where each node has zero or more child nodes, except for the root node which has no parent.

2. What are the basic components of a tree structure?

Ans: The basic components of a tree structure are nodes, edges, root node, and leaf nodes.

3. What is a root node?

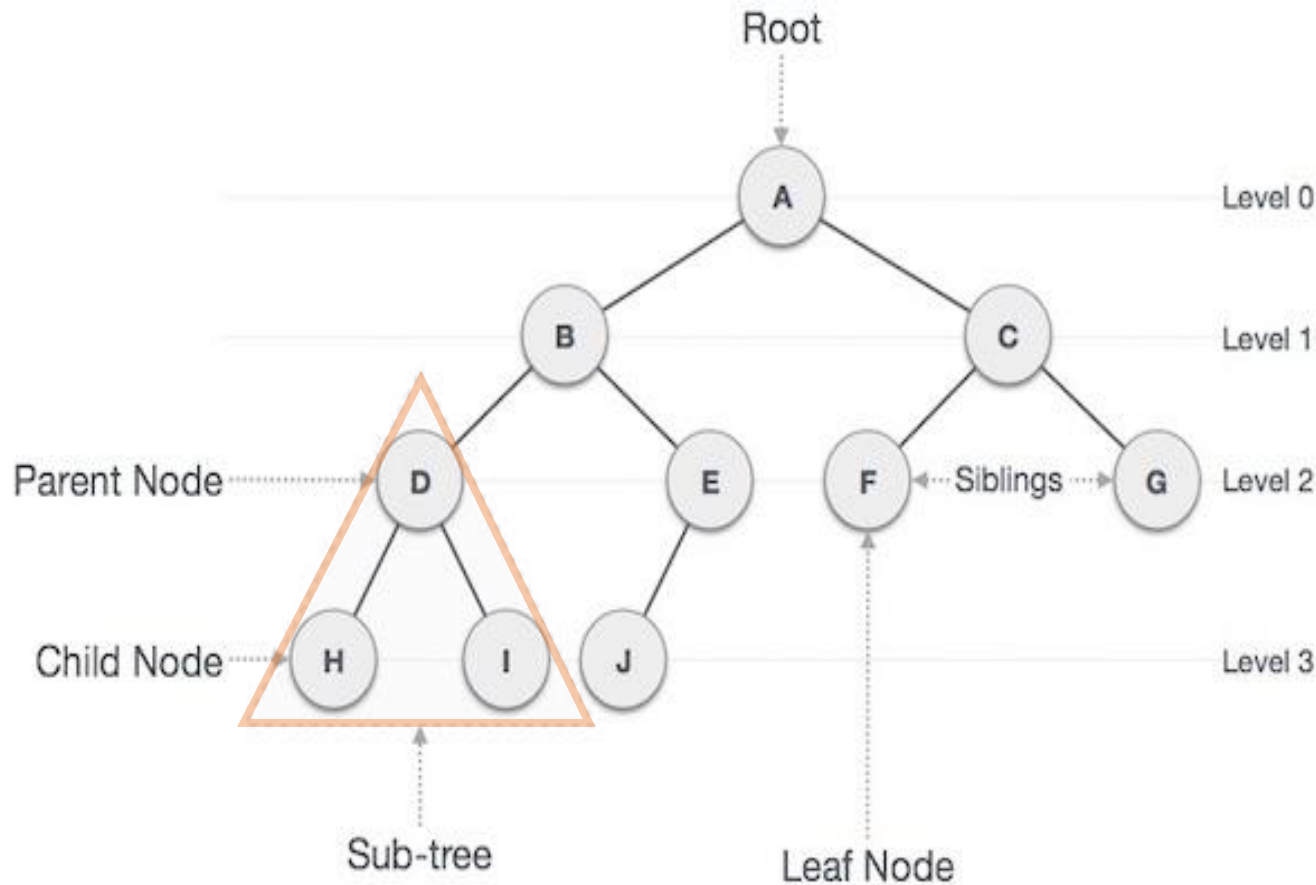
Ans: The root node is the topmost node in a tree structure, and it has no parent nodes.

4. What are leaf nodes?

Ans: Leaf nodes are the nodes in a tree structure that have no child nodes.

Tree Data Structure

Basic components of a tree?



Depth: The depth(or level) of a node is its distance(i.e. no of edges) from tree's root node.

For example:

Depth of the node D is 2

Depth of the node I is 3

Height: The height is number of edges between root node and furthest leaf.

For example:

Height of the tree is 3.

Cousins: Cousins of the node F and G are D and E in the tree.

Tree Data Structure

Basic components of a tree?

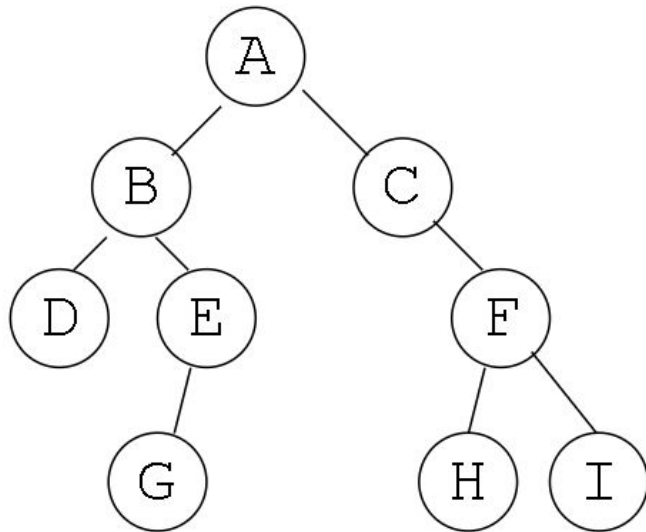
level

0

1

2

3



root: A

node: A, B, C, ..., H, I

father of B: A

sons of B: D, E

left son of B: D

right son of B: E

depth: 3

ancestors of E: A, B

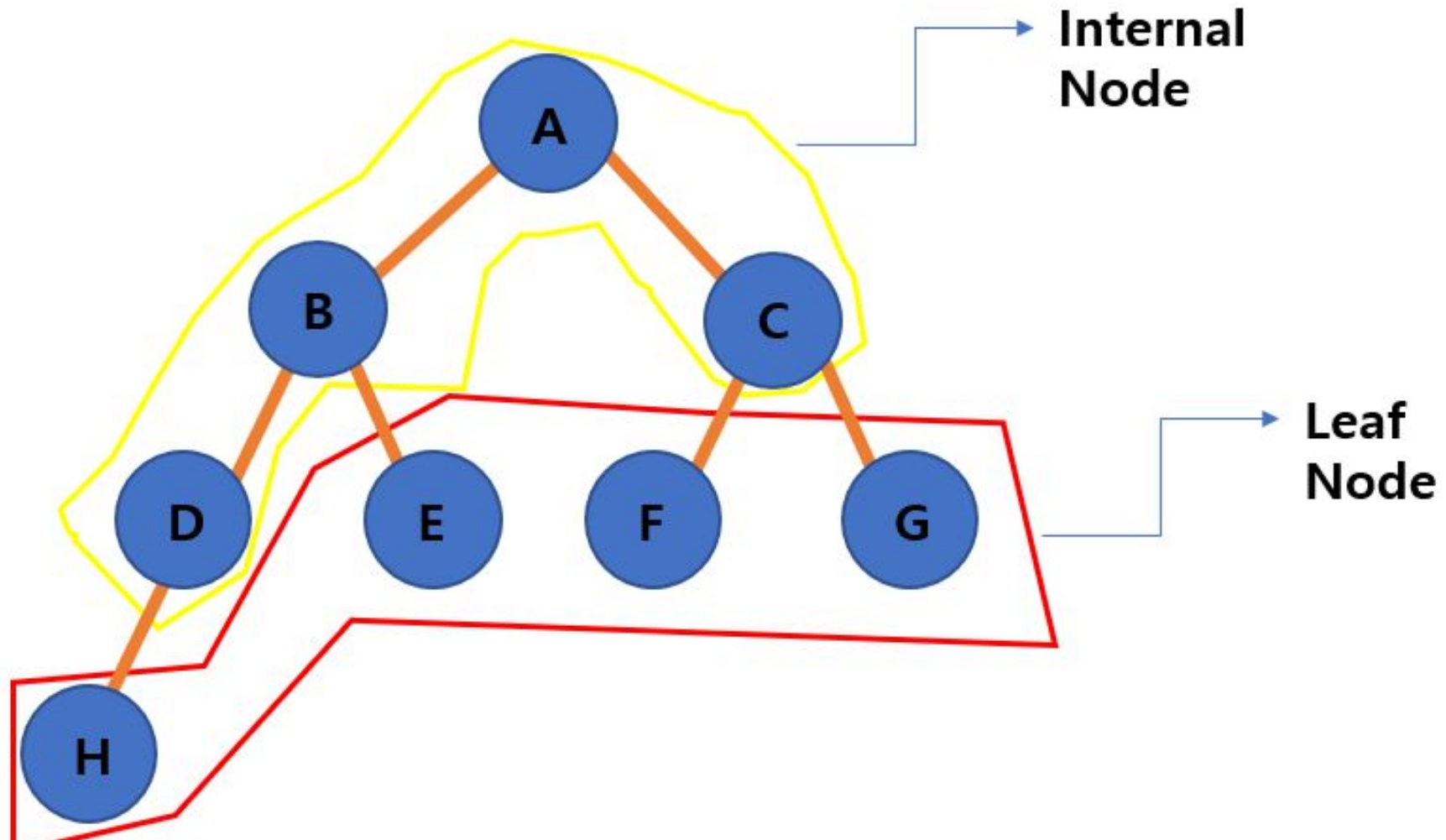
descendants of B: D, E, G

Ancestor of a Node: Any predecessor nodes on the path of the root to that node are called Ancestors of that node.

Descendants of a Node: All nodes under a node (A) is the descendants of the node (A).
For example:
Descendants of C is F,H,I.

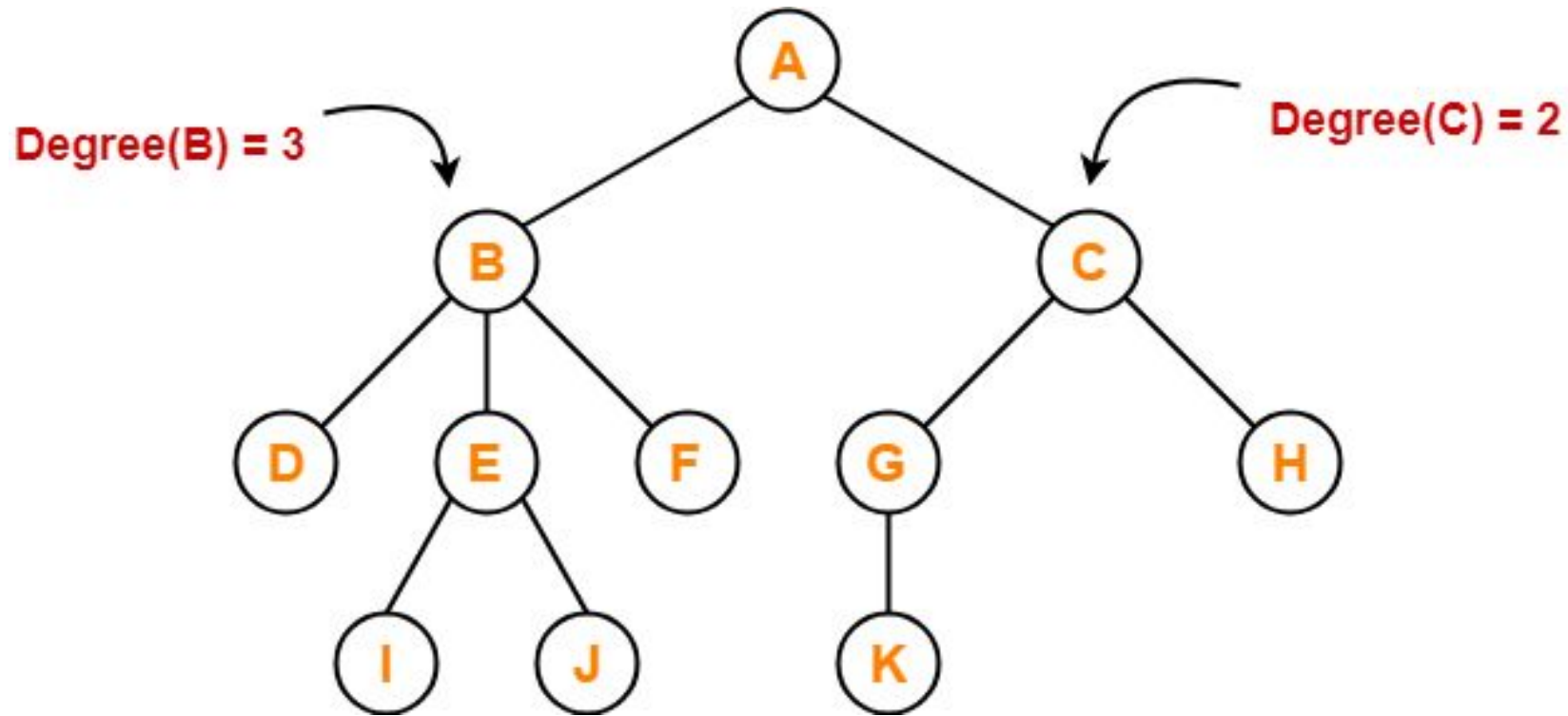
Tree Data Structure

Basic components of a tree?



Tree Data Structure

Basic components of a tree?



Here,

- Degree of node A = 2
- Degree of node B = 3
- Degree of node C = 2
- Degree of node D = 0
- Degree of node E = 2
- Degree of node F = 0
- Degree of node G = 1
- Degree of node H = 0
- Degree of node I = 0
- Degree of node J = 0
- Degree of node K = 0

Tree Data Structure

1. What is a traversal of a tree?

Ans: Traversal of a tree is the process of visiting each node in the tree exactly once.

2. What are the three common methods for tree traversal?

Ans: The three common methods for tree traversal are pre-order traversal, in-order traversal, and post-order traversal.

3. What is pre-order traversal?

Ans: Pre-order traversal is a tree traversal method where the root node is visited first, followed by the left subtree and then the right subtree. Root $\square$ Left $\square$ Right

4. What is in-order traversal?

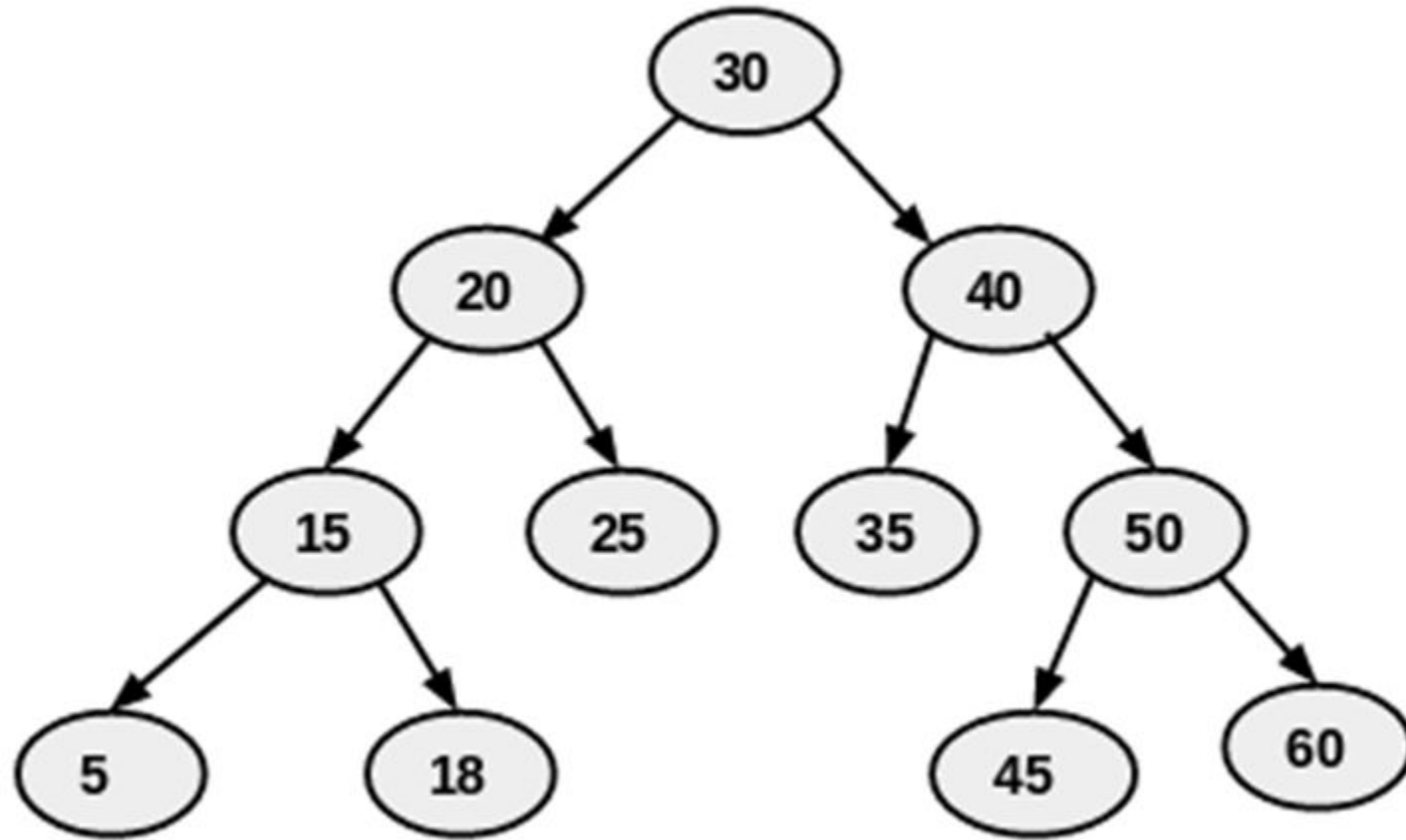
Ans: In-order traversal is a tree traversal method where the left subtree is visited first, followed by the root node and then the right subtree. Left $\square$ Root $\square$ Right

5. What is post-order traversal?

Ans: Post-order traversal is a tree traversal method where the left subtree is visited first, followed by the right subtree, and then the root node. Left $\square$ Right $\square$ Root

Tree Traversal

Example of pre-order, in-order, and post-order traversals



Pre-order (Root □ Left □ Right)

{30 , 20 , 15 , 5 , 18 , 25 , 40 , 35 , 50 , 45 , 60}

In-order (Left □ Root □ Right)

{5 , 15 , 18 , 20 , 25 , 30 , 35 , 40 , 45 , 50 , 60}

Post-order (Left □ Right □ Root)

{5 , 18 , 15 , 25 , 20 , 35 , 45 , 60 , 50 , 40 , 30}

Thank You